Financial Analysis Project using Excel

Analysis of Loans and Investments using Financial Functions

Introduction

- Financial analysis helps in evaluating loans and investment decisions
- Excel provides built-in financial functions for accurate analysis
- This project demonstrates real-life financial scenarios using Excel formulas

Project Scope

- Use Excel for financial calculations
- Analyze loans, EMIs, and investments
- Apply financial functions like PV, NPV, IRR, etc.
- Draw conclusions based on numerical results

Project Objectives

- Understand key financial concepts
- Learn Excel financial functions
- Compare different financial options
- Improve decision-making using financial analysis

Key Financial Concepts

- Annuity
- Present Value (PV)
- Net Present Value (NPV & XNPV)
- EMI and Loan Analysis
- IRR, XIRR, MIRR

Annuity

An annuity is a series of equal payments made at regular intervals

Examples:

- EMI payments
- Insurance premiums
- Pension payments

In Excel:

- Cash paid → negative value
- Cash received → positive value

Present Value (PV)

PV shows the current value of future payments

=PV(rate, nper, pmt, [fv], [type])

rate: Interest rate per period.
nper: Total number of payment periods.
pmt: Payment made each period.
fv: Future value (optional).
type: Payment type (0 for end of period, 1 for beginning of period).

PV Example

(Refrigerator Case)

- Price: ₹32,000
- Interest Rate: 13%
- Years: 8
- Annual Payment: ₹6,000

Result:

- Paying yearly at end of year has **lower present value**
- Hence, it is the **better option.**

1		
2	Price	32000
3	Interest Rate	0.13
4	No. of payments	8
5	Payment	-6000
6	Payments at end of the year	
7	PV	$ 28,792.62

Net Present Value (NPV)

Meaning:

- NPV calculates profitability of an investment

Decision Rule:

- NPV > 0 → Accept investment
- NPV < 0 → Reject investment
- Formula =NPV(rate, cashflows)

Investment Comparison using NPV

- Two investments may have similar cash inflows
- NPV helps identify which gives higher value today

48	Decision on Investments		
49			
50	Interest Rate	0.2	
51		Cash Flow	
52	Time	Investement 1	Investement 2
53	1	-10000	-5000
54	2	25000	20000
55	3	-7000	-8000
56	Total	8000	7000
57			
58	NPV (End Year)	₹ 4,976.85	₹ 5,092.59
59	NPV (Beginning Year)	5972.22	6111.11
60	NPV (Middle Year)	₹ 5,451.87	₹ 5,578.66

XNPV (Irregular Cash Flows)

- Used when cash flows occur on different dates
- More realistic for real projects

- =XNPV(rate, values, dates)

EMI (Loan Analysis)

Meaning:

- EMI is a fixed monthly payment for a loan

- =PMT(rate, nper, pv)

EMI Example (Home Loan)

- Loan Amount: ₹50,00,000
- Interest Rate: 11.5%
- Tenure: 25 years

Result:

- Excel calculates monthly EMI.

EMI

Rate per annum	0.12
Rate per month	0.01
Term	25
No. of monthly payments	300
Loan Amount (PV)	5000000
FV	0
Type	1
EMI	-₹52,139.81

Rate per annum	0.16
Rate per month	0.013333333
Term	8
Loan Amount (PV)	100000
FV	0
Type	0
EMI	-₹13,261.59

Rate per month	0.013	
No. of monthly payment	8	
Loan Amount (PV)	100000	
FV	0	
Type	0	
EMI		(₹13,242.27)

Interest paid between 2nd and 3rd months
2132.23
₹ 2,132.23
Principal paid between 2nd and 3rd months
24352.30
₹ 24,352.30

Interest & Principal Components

EMI consists of:

- Interest component
- Principal repayment

Excel functions:

- =IPMT() → Interest
- =PPMT() → Principal

Cumulative Interest & Principal

- Used to find:
  - Interest paid between two months
  - Principal paid between two periods

  EXCEL FUNCTIONS:
  - =CUMIPMT()
  - =CUMPRINC()

RATE (Interest Rate Calculation)

Purpose:

- Finds interest rate based on EMI and tenure

EXCEL FUNCTIONS:

- =RATE(nper, pmt, pv)

Example Result:

- Interest Rate = 8%

NPER (Loan Duration)

Purpose:

- Calculates number of payments required to clear loan

- EXCEL FUNCTIONS:

  =NPER(rate, pmt, pv)

Example Result:

- Loan cleared in 12 months

IRR (Internal Rate of Return)

- IRR is the rate where NPV = 0
- Used to evaluate investment profitability

Excel Formula:
=IRR(values)

Problems with IRR

- IRR may:
  - Give multiple values
  - Not exist
  - Mislead for projects of different sizes

  In such cases, **NPV is more reliable**

XIRR & MIRR

XIRR:

- Used for irregular cash flows
- =XIRR(values, dates)

MIRR:

- Considers different finance & reinvestment rates
- =MIRR(values, finance_rate, reinvest_rate)

Key Insights

- Excel simplifies financial analysis
- PV & NPV help compare options
- EMI analysis explains loan structure
- NPV is better than IRR in many cases

Real-Life Applications

- Loan planning
- Investment comparison
- Home loan decisions
- Business project evaluation

Tools Used

- Microsoft Excel
- PowerPoint / PDF

Conclusion

- Financial functions improve decision-making
- Excel provides accurate and fast analysis
- This project builds strong financial fundamentals